

AR-based Interactive Bookstore Navigation

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Abstract— This project will create an augmented reality (AR)-based navigation system for bookstores using data structures like stacks, queues, and trees. Users can interact with an AR interface to find and navigate through a bookstore efficiently. The system will prioritize user preferences, display real-time updates, and track the shortest paths to bookshelves.

Keywords—stacks, queue, trees, AR, bookstore navigation.

I. INTRODUCTION

This project will create an augmented reality (AR)-based navigation system for bookstores using data structures like stacks, queues, and trees. Users will interact with an AR interface to find and navigate through a bookstore efficiently. The system will prioritize user preferences, display real-time updates, and track the shortest paths to bookshelves, enhancing the browsing experience.

This project involves the development of a Graphical User Interface (GUI) system designed to streamline and simplify the interactions between administrators and users within an application environment. The system provides distinct interfaces tailored to the needs of two user roles—admin and standard user—enhancing both usability and functionality for each role.

II. LITERATURE SURVEY

The paper "*Augmented Reality Navigation*" shows that Augmented Reality (AR) can make navigating spaces much easier and more engaging by displaying useful information directly onto real-world places. This overlay of information helps users understand their surroundings better and find their way more easily. The paper highlights that combining AR

with backend systems, where data is stored and quickly updated, allows users to access the latest information, making navigation more interactive and responsive to their needs.

AR is useful in many fields, such as retail, healthcare, and education, where it can make finding things easier and more intuitive. A key takeaway from the paper is the importance of designing AR apps with users in mind so that they are easy to use and understand.

Using these ideas, we try to designed a navigation system for a bookstore using AR. We incorporated data structures like trees, stacks, and queues to make it fast and efficient. Trees help organize book sections and categories, stacks keep track of recent locations, and queues help manage multiple user requests at once. This approach creates a smooth and enjoyable navigation experience for users looking for books in the store.

Augmented Reality Navigation paper - https://www.researchgate.net/publication/342383348_Augmented_Reality_Navigation

III. METHODOLOGY

Starting the Project

Define a Book class to store information about each book, including its reference number, title, author, and whether it's trending or not.

Use a BookBST class to store and organize books by reference number. This structure helps make searches, additions, and deletions fast.

Managing Data

Set up functions to load book data from an Excel file at startup and save any updates back to the file. This way, data stays up-to-date every time the system is used.

Include a feature to mark certain books as trending, making it easy to highlight popular books.

Designing the AR Layout

Create a virtual lab layout where users can see book locations in an augmented reality (AR) setting. This layout guides users in finding books and shows where trending books are located.

Assign each book a virtual spot in the AR layout with visual markers, helping users locate specific books.

Integrate this AR layout with the Tkinter interface by creating a clickable mini-map that shows the virtual layout, trending books, and guides users to search results.

User Interface (UI) Design

Set up a main window in Tkinter with tabs for different roles: admin for managing books and users for searching and exploring the AR layout.

The separate tabs for admin and user tasks allow the admin to manage book data, while users can focus on using the AR map to find books.

Admin Features

In the admin tab, admins can add new books by entering details like title and reference number. The book is then added to the BST and displayed in the main list.

Admins can delete books, updating both the BST and the Excel file to keep records accurate.

Admins can mark books as trending, making them stand out in the main list.

A search option lets admins find specific books and place markers on the AR map to show the book's virtual location.

User Features

Users can search for books using their reference numbers. The system will then display the book's details and its virtual location in the AR layout.

Users can view a list of trending books and locate them on the AR map.

Past searches are tracked in a list, allowing users to revisit previous searches and find the corresponding AR markers.

Running the System

Launch the program by starting the Tkinter main window. Actions taken by users update the AR navigation in real time.

Users can switch between the admin and user tabs easily to perform different tasks.

Methodology Summary

The BST structure ensures efficient handling of large numbers of books by making search, insertion, and deletion quick and manageable.

The system has clear access control, with specific tasks available only to admins or users.

All book data, search history, and AR markers are saved in Excel files, making it easy to maintain data between sessions.

The AR layout and Tkinter interface together provide an interactive experience for users, allowing them to locate both trending and specific books visually and easily.

IV. RESULTS

The project, "Augmented Reality (AR)-Based Interactive Bookstore Navigation Using Stacks, Queues, and Trees," achieved significant progress in backend data structure integration and GUI development for admin and user roles, with the following outcomes:

1. Backend Code for Efficient Bookstore Navigation:

- A tree structure in the backend effectively represents the bookstore layout, enabling organized book search and guided navigation. Although the AR component wasn't fully implemented, the foundational layout was designed based on the VLSI-UG lab, laying groundwork for future AR expansion.

2. Prioritization of Popular Books:

- Using a stack structure, trending and high-demand books are placed at the top of search results, ensuring prominent visibility. This prioritization logic is seamlessly integrated

within the GUI, allowing users to quickly locate popular titles.

3. Queue Management for Customer Assistance:

- The backend implements a queue system for handling customer assistance requests on a first-come, first-served basis. Real-time updates are displayed in the GUI, improving the user experience by keeping wait times visible.

4. Admin and User-Friendly GUI:

- The GUI was developed for both admin and user roles, providing a clear, intuitive interface for backend interactions. Admins have streamlined access to manage inventory and user requests, while users benefit from a smooth, visually organized browsing and searching experience.

Although the AR component is limited to a layout based on the VLSI-UG lab, the project successfully integrates data structures with a functional GUI, setting the stage for a more interactive and efficient bookstore experience.

V. CONCLUSION:

The "Augmented Reality (AR)-Based Interactive Bookstore Navigation Using Stacks, Queues, and Trees" project successfully combines AR, backend data structures, and a user-friendly GUI to redefine bookstore navigation and interaction. By leveraging data structures like trees for bookstore layout, stacks for trending books, and queues for customer assistance, this project demonstrates how traditional data structures can be applied innovatively in real-world applications. The result is an efficient, interactive, and enjoyable browsing experience for users, making it easier to locate books, receive timely assistance, and discover new titles based on personal preferences. Overall, this system enhances user engagement, reduces search time, and provides a modernized bookstore experience.

VI. FUTURE SCOPE

1. Better Book Recommendations:

- Adding machine learning could help the system learn what each user likes and make smarter book recommendations based on their past choices and interests.

2. Support for Multiple Stores:

- Expanding the system to include multiple stores would let users check for book availability across different locations and find the nearest store with the book they want.

3. Voice and Gesture Controls:

- Adding voice commands and hand gestures would make the system even easier to use, especially within the AR experience, allowing users to navigate without touching the screen.

4. Virtual Bookshelf in AR:

- Users could browse virtual shelves, read summaries, and see reviews directly through the AR interface, making the experience more interactive and fun.

5. Helpful Store Insights:

- Bookstore owners could use data from the system to see which books are popular and adjust inventory or promotions based on customer interests.

6. Access on Different Devices:

- Making the system available on various devices (like smartphones, tablets, and computers) would let more people use it, whether they're at home or in-store.

- Expanding to multiple store locations would allow users to check book availability across different stores and locate the nearest store with the book they want.

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